

The Code-Mixing Penalty and the Reference Problem: A Measurement Study of Hinglish Clinical Retrieval-Augmented Generation

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Abstract

Patients in multilingual communities routinely describe symptoms in romanised, code-switched language, while the clinical literature that could answer them is written in formal English. We present a measurement study of retrieval-augmented generation (RAG) for Hinglish patient questions grounded in English clinical case reports, built on 3,015 query–case pairs that link the MMCQS question set to the MultiCaRe corpus through a label-leakage gate. Four pre-specified null hypotheses are tested. Code-mixing imposes a retrieval penalty that is positive in all 28 system–metric combinations tested and significant in 26, roughly 4.4 times larger for lexical retrieval than for dense cross-lingual embedding at rank 1 and wider at depth: BM25 is the best system on English queries and the worst on Hinglish. MuRIL, pretrained on Devanagari Hindi, retrieves at chance on romanised input, and transliterating the queries into Devanagari recovers only 6.5% of its gap; translating them into English helps dense retrieval but harms lexical retrieval, because the English arm of the comparison is a clinician-written summary rather than a translation. Scored against the evidence it was given, grounding significantly increases concept precision; scored against a reference neither arm saw, the gain shrinks or disappears, so it is reported as a reference-dependent effect rather than a general improvement in answer quality. Grounded factual support is stable across code-mixing intensity while the ungrounded arm degrades. Evidence provenance significantly changes refusal behaviour (Cochran’s $Q = 9.09$, $p = 0.028$), but its effect on answer quality could not be tested adequately and remains provisional. All the corrections, negative results, and the evaluation hazards are given in full.

Keywords: Hinglish, code-mixing, retrieval-augmented generation, clinical question answering, cross-lingual retrieval, evaluation methodology.

1. Introduction

Problem. A patient types: “Doctor, meri beti ko skin par rash ho gaya hai, aur khujli ho rahi hai.” The case report that could give an answer is: “erythematous maculopapular rash with pruritus.” The clinical content is included in the evidence, but the surface form of the question is not what the system expects.

In fact, many patients describe their symptoms online in a register that is a mix of Hindi and English, commonly referred to as “Hinglish” (Hindi-English), and clinical literature is written in formal English with most retrieval pipelines being developed and tested using only monolingual queries.

Retrieval-augmented generation (RAG) holds the potential to solve this: retrieve a case in English that matches the patient's circumstances, then generate an answer **grounded in that case, in the patient's own language**. The question of whether that promise will hold true for code-mixed data is an empirical one, with two distinct components: is code-mixed surface form hindering retrieval, and does grounding improve the factual quality of the generated answer?

Gap. Neither component is isolated in existing evaluations. First, code-mixed queries are rarely compared against gold English versions of the same question, so the retrieval cost of code-mixing is confounded with topic. Second, grounding studies regularly score answers against the very evidence on which the grounded arm was conditioned, a reference the ungrounded arm never saw. Third, grounding and code-mixing robustness are seldom tested as hypotheses fixed in advance, or replicated across generator families.

Research questions. RQ1: how much does romanised code-mixing cost at the retrieval stage, and does the cost depend on the retrieval method (H04)? RQ2: does grounding change factual support, and is that change robust to code-mixing intensity (H01, H02)? RQ3: does the provenance of the evidence matter (H03)?

Approach. This paper is a measurement paper and not a system-improvement paper. We build a Hinglish clinical RAG pipeline purposefully from standard components (3,015 Hinglish patient questions from MMCQS [1] with case reports from MultiCaRe [2] together with embedding over a FAISS index and grounded prompting of an instruction tuned generator), so as to reflect the setting and not the particular architecture. The design choices are justified by what each isolates: LaBSE [3] because cross-lingual alignment is its training objective; BM25 [12] and reciprocal-rank fusion [13] as the standard lexical baseline and a parameter-free way of combining the two signals; paired Hinglish–English queries because they hold content fixed while the surface form varies; and two scoring references, one the grounded arm saw and one neither arm saw, because the difference between them is the circularity this paper measures.

The four null hypotheses tested are: (H01) grounding does not influence factual **support**; (H02) the intensity of code-mixing has no effect on the factual difference between grounded and ungrounded; (H03) authoritative case evidence does not significantly influence the factual correctness over general biomedical text; and (H04) the retrieval quality is not different from code-mixed query as compared to the English equivalent. H01–H03 were specified in the project proposal and public repository before any data were analysed (commit 0917322, 9 February 2026); H04 was formulated after the initial retrieval experiments and is reported as a post-hoc hypothesis, tested confirmatorily on the full matched set (Sect. 4).

Not all of the project's original claims held up to scrutiny, and we report these corrections as findings in their own right: a code-mixing index that misclassified common English words, an index configuration that silently truncated every case narrative, a published context-selection heuristic that never fires on this data, and an evaluation protocol whose circular reference inflates the apparent benefit of grounding. Rejecting a null hypothesis is the positive outcome; it means the corresponding effect is real.

Contributions. The study makes three contributions. First, it quantifies the retrieval-stage code-mixing penalty, measured against gold **English summaries** of the same questions behind a label-leakage gate, together with the lexical/dense asymmetry and its script-mismatch mechanism: the penalty is positive in all 28 system–metric combinations, 4.4 times larger for lexical than for dense retrieval at rank 1, transliteration recovers only 6.5% of MuRIL's gap, and translation helps dense but not lexical retrieval (Sect. 5.1). Second, it provides **pre-specified hypothesis tests for the grounding effect and for code-mixing robustness, reported under the tests that were planned and replicated across two generator families under oracle evidence** (Sects. 5.3 and 5.4). Third, it demonstrates that evaluation design changes the reported benefit of grounding — **positive in five of five conditions against the evidence reference and in one of five against an unbiased one** — and supplies the degenerate baselines **and the extractor audit** required to detect such artefacts (Sects. 5.2 and 5.3).

2. Related Work and Research Gap

Work on code-mixed clinical language has concentrated on the query side of the encounter. MedSumm [1] introduced the MMCQS corpus, pairing Hinglish patient questions with human-written English summaries and textual image descriptions, and proposed multimodal summarisation of those queries; HealthAlignSumm [4] extended this line with alignment-based training for multimodal summarisation of code-mixed healthcare dialogues. The present study uses the MMCQS questions and gold English summaries as matched query pairs and the image descriptions only as a scoring reference; no images are processed at any stage, and the task studied is retrieval and grounded generation rather than summarisation.

A second body of work develops retrieval-augmented generation for medicine. Xiong et al. [5] benchmark medical RAG across corpora, retrievers and generators; MMed-RAG [6] adds domain-aware retrieval and adaptive context selection for medical vision-language models; and factuality-targeted augmentation has been applied to radiology reporting [7] and to clinical pipelines such as AlzheimerRAG [8]. Vision-language assistants such as LLaVA-Med [9] generate without a retrieval stage.

Across this line the query language is uniformly English, and reported grounding benefits are frequently computed against the retrieved evidence itself, a reference the grounded arm was conditioned on and the ungrounded arm never saw. Section 5.3 quantifies how much of the apparent benefit this circularity contributes in our setting, and Sect. 5.6 tests whether the adaptive context-selection heuristic of [6] transfers to it.

Code-switching has also become an evaluation axis in its own right outside medicine. CroCoSum [10] benchmarks cross-lingual code-switched summarisation, and HiFactMix [11] pairs a Hinglish benchmark with a graph-aware model for political claim verification; neither addresses the clinical domain or isolates the retrieval stage.

On the retrieval side, lexical rankers such as BM25 [12] depend on surface-form overlap, which romanisation removes; dense cross-lingual encoders such as LaBSE [3] place translations near one another in a shared space, and reciprocal-rank fusion [13] combines the two signals; encoders pretrained on Indian languages, notably MuRIL [14], target Devanagari Hindi, raising the script-mismatch question that Sect. 5.1 answers empirically. One boundary is explicit: the present system is text-only. Earlier project documents mentioned multimodal components, but none was implemented or evaluated here and no multimodal claim is made.

To our knowledge, no prior study of code-mixed clinical question answering (i) isolates the retrieval-stage penalty of romanisation by comparing code-mixed queries against gold English versions of the same questions under a label-leakage gate, (ii) tests grounding and code-mixing robustness as pre-specified hypotheses across generator families, or (iii) measures how strongly the choice of scoring reference changes the reported grounding benefit. This paper does all three; Table 1 positions it against the closest prior work.

Table 1 Positioning against the closest prior work. ✓ = addressed; — = not addressed; ~ = partly. ‘Gold English pairs’ = the same question available in English for a matched comparison; ‘reference bias checked’ = the grounding benefit scored against a reference the grounded arm did not see.

Work	Code-mixed queries	Clinical domain	Retrieval stage isolated	Gold English pairs	Reference bias checked
MedSumm [1]	✓	✓	—	✓ (summaries)	—
HealthAlignSumm [4]	✓	✓	—	—	—
Xiong et al. [5]	—	✓	✓	—	—
MMed-RAG [6]	—	✓	~	—	—
CroCoSum [10]	✓	—	—	✓	—
HiFactMix [11]	✓	—	~	—	—
This work	✓	✓	✓	✓ (summaries)	✓

3. Materials and Methods

The pipeline is intentionally standard; every design choice that turned out to matter (chunking, sequence limits, the code-mixing index, the scoring reference) is reported with its measured consequence in Sect. 5. All symbols are defined at first use.

3.1 Problem Formulation

The evaluation items are pairs (q, q_E) : a Hinglish patient question q together with its gold, human-written English summary q_E — a clinician-style condensation of the same question (median 20 words against 92 for q ; Sect. 5.1 measures the consequence). The evidence corpus comprises 10,000 English MultiCaRe case narratives drawn from 18 condition groups, segmented into 41,746 passages. A retrieval system s maps a query onto a ranked list of cases; $\text{Success@1}_s(\cdot)$ is the proportion of queries for which the top-ranked case falls in the same condition group as the query (a hit rate: with a median of 597 relevant cases per query it is not recall, and it is named accordingly). This condition-group criterion is deliberately coarse (Sect. 6.1), but it is the same across both languages, which is exactly what the penalty estimand requires. The per-system code-mixing penalty is

$$\Delta_s = \text{Success@1}_s(q_E) - \text{Success@1}_s(q) \quad (1)$$

where a positive Δ_s indicates that the deployed, code-mixed form retrieves worse than its English equivalent. An instruction-tuned generator g produces a zero-shot answer $a_0 = g(q)$ and a grounded answer $a_r = g(q, c)$ conditioned on a retrieved or oracle case c ; both answers are in Hinglish. The answers are then scored by overlap with a fixed 26-concept clinical lexicon against a designated reference text: factual support is concept precision (the share of concepts the answer asserts that are actually present in the reference), concept recall is the share of reference concepts the answer asserts, and F1 combines the two. The estimands are the per-system penalty Δ_s , paired within-query differences between arms scored identically, the change in that paired difference across levels of the Hindi proportion of q (with per-arm correlations as its decomposition), and contrasts across matched evidence corpora.

3.2 Architecture

The deployed path embeds the Hinglish query with LaBSE [3] and searches a FAISS index of the 41,746 passages; passage scores are aggregated to cases and the top case is returned. Lexical baselines (TF-IDF and BM25 [12]) rank the full documents, and a hybrid system fuses the dense and BM25 rankings by reciprocal-rank fusion [13]. The retrieved case is injected into a grounded prompt instructing the generator to answer the patient in Hinglish using only the supplied evidence; the zero-shot arm receives the identical instruction without evidence. An oracle-evidence condition supplies a case filtered to the query’s own condition group, upper-bounding what retrieval could deliver; the real-retrieval condition uses the deployed path unchanged. The evaluation harness extracts lexicon concepts from each answer and reference and computes factual support, recall, F_1 , and refusal status, to which the statistical procedures of Sect. 4.2 are applied.

4. Experimental Design and Evaluation

The study tests four pre-specified null hypotheses. H01–H03 were fixed in the project proposal and the public repository before any analysis was run (commit 0917322, 9 February 2026; <https://github.com/saaachis/hinglish-grounded-medical-rag/commit/0917322>), and the first prototype followed on 4 March 2026. H04 was added after the initial retrieval experiments and is tested confirmatorily on the full matched set. This is a timestamped public record rather than a registry entry, and the deviations from the planned analysis are listed at the end of Sect. 4.2. Rejecting a null is the positive result: it indicates that the corresponding effect is real. And the two-sided tests are conducted throughout, with false-discovery-rate correction applied within each hypothesis family (Benjamini–Hochberg, BH); cluster bootstraps deal with the non-independence introduced by repeated reference texts (Sect. 6.1).

4.1 Datasets and Preprocessing

The queries come from MMCQS [1], a large repository of Hinglish patient questions, each accompanied by a human-written English summary and a textual image description; only these text fields are used in this study, and no images are processed. The summary is a clinician-style condensation of the question (median 20 words against 92 for the Hinglish query), not a translation; Sect. 5.1 measures what that difference contributes to the penalty. Evidence comes from MultiCaRe [2], an open-access collection of clinical case reports: 61,316 cases pass filtering, of which 10,000 are indexed, balanced across 18 condition groups. The median case runs 554 words, and passage chunking yields 41,746 passages. Pairing then produces 3,015 query–case evaluation pairs at 100% query coverage.

The choice of evidence corpus is itself a finding. An initial attempt to match MMCQS to Open-i radiology reports by TF-IDF similarity produced only 11 usable pairs: the corpora share almost no topical overlap.

Switching the evidence base to MultiCaRe and matching with LaBSE similarity plus condition-aware filtering produced 3,015 pairs, a 274-fold increase. Pairing decisions of this kind can dominate everything downstream and deserve explicit reporting.

A leakage gate is required because MMCQS’s English summary embeds an image caption that names the query’s condition group in 96.2% of rows and reproduces the underscore-joined group label exactly in 33.6%; that is, the summary itself contains the relevance label. Any English-versus-Hinglish comparison that uses the full summary is therefore inflated by construction. All English-side queries use the caption-stripped summary, and an automated gate asserts that no label string survives.

4.2 Baselines and Evaluation Metrics

Four retrieval systems are compared on the passage-chunked index: TF-IDF, BM25 [12], dense LaBSE retrieval [3], and a hybrid fusing the BM25 and LaBSE rankings by reciprocal-rank fusion [13]. A supplementary single-vector index (one embedding per case) additionally hosts MuRIL [14] and multilingual-e5-base for the script-mismatch analysis. Two query-side remedies are tested as further baselines: transliterating the romanised query into Devanagari for MuRIL (a hand-verified lexicon covering 87% of Hindi token occurrences plus rule-based mapping for the remainder; a trained transliterator could not be installed in the environment, so the mapping is a lower bound), and translating it into English with gpt-oss-120b before retrieval, on a stratified 500-query subset. The primary metric is Success@1 against condition-group relevance, the proportion of queries whose top-ranked case falls in the query’s condition group; Success@k for $k \in \{3, 5, 10\}$, MRR@10 and nDCG@10 are also reported. nDCG uses the corpus-level ideal ranking (the min(relevant, k) hits placed first), and a graded variant awards grade 2 to a same-group case that also shares at least one lexicon concept with the gold English question: an automatic proxy for finer relevance, not a clinical judgment. The random floor of Success@1 for condition-blind retrieval is 0.0626, rising to 0.4739 at rank 10. Because the relevance criterion is coarse, absolute Success@1 is a lower bound on true relevance (Sect. 6.1). The penalty Δs is tested per system and per metric, with McNemar’s test on paired top-k hits and the Wilcoxon signed-rank test on MRR and nDCG: a penalty appearing under one configuration is a property of that configuration, whereas one holding across methods is a property of code-mixing itself.

For generation, each query is answered zero-shot and grounded (Sect. 3.2). Answers are scored with the 26-concept lexicon against two references: the retrieved evidence (*circular*, because the grounded arm was conditioned on it and the zero-shot arm never saw it) and the dataset’s image description, which neither arm saw (*unbiased*).

Factual support equals concept precision, and the commonly reported “hallucination rate” is exactly $1 - \text{precision}$ (verified $\min = \max = 1.0$ across rows), so hallucination and factual support are one number, not two, and are never reported here as separate findings. Since precision has no recall term, degenerate constant answers can outscore real systems (Sect. 5.2); F_1 is thus reported when quality is at issue, degenerate baselines ship with the scored tables, and refusal and coverage (answers asserting at least one scoreable concept) appear beside each factuality number.

The statistical toolkit includes paired within-query deltas with Cohen’s d, McNemar’s test for paired binary outcomes, and Cochran’s Q for refusal across matched evidence conditions, together with cluster bootstraps over repeated references, BH correction within hypothesis families, one-way ANOVA and Kruskal–Wallis on the paired gain across code-mixing tertiles for the pre-specified H02 test, and Spearman ρ with 95% confidence intervals for dose–response.

Deviations from the pre-specified plan. Five departures from the original analysis plan are disclosed. (1) H02 was planned as a two-way ANOVA / Kruskal–Wallis on code-mixing level; it is reported that way as the primary test, with the per-arm correlations as an exploratory decomposition. (2) Factual consistency and hallucination rate were planned as two outcomes; hallucination is exactly $1 - \text{precision}$ and is not reported separately. (3) Concept precision was the planned quality metric; F_1 is reported wherever quality is at issue, because degenerate baselines showed that precision’s optimum is a one-word answer. (4) The Code-Mixing Index was planned over a fixed Hindi word list; that list counted “doctor” and “please” as Hindi and was repaired. (5) H03 was planned on answer quality; it is reported as exploratory, with refusal as a secondary outcome.

4.3 Implementation Details

The pipeline uses LaBSE sentence embeddings with a FAISS index over the 41,746 passages; lexical baselines index full documents. Generators are llama-3.1-8b-instant, a hosted instruction-tuned model, and gpt-oss-120b, an open-weight model; both receive identical prompts. Midway through the study the hosting provider decommissioned llama-3.1-8b-instant, which now returns HTTP 404: its archived outputs remain re-scorable, but no new generations are possible (Sect. 6.1). An early index build imposed a 128-token sequence limit whose consequences are reported as a corrected result in Sect. 5.6; all dense results elsewhere use the corrected, passage-chunked index unless stated otherwise.

Experiments ran under Python 3.12 with sentence-transformers 3.0.1, transformers 4.42.4, PyTorch 2.3.1, faiss-cpu 1.8.0, scikit-learn 1.8.0, NumPy 1.26.4, pandas 2.2.1, SciPy 1.12.0 and matplotlib 3.8.3, with rank_bm25 for the BM25 baseline; full pinned versions are listed in requirements.txt in the released repository. The translate-then-retrieve baseline uses the hosted openai/gpt-oss-120b as translator, and the extractor audit of Sect. 5.2 uses qwen/qwen3.8-27b as an independent reader, a model family used nowhere else in the study.

All stages ran on commodity CPU hardware (a four-core Intel Core i3-8130U, 12 GB memory); no GPU was used at any stage. Approximately 7,200 generation calls were issued in total: about 4,300 on llama-3.1-8b-instant and about 2,800 on the gpt-oss models. Random seeds are fixed at 42 throughout. Code and data availability are stated in the Declarations.

5. Results

Subsections follow the pre-specified order: the retrieval-stage hypothesis H04 first, a validity audit of the scoring instrument before any generation numbers are presented, then H01, H02 and H03, and finally the negative and corrected results. H04 is rejected for every system and metric; H01 is rejected under the evidence-referenced protocol; the pre-specified H02 test is borderline (Sect. 5.4); H03 is exploratory (Sect. 5.5). Bold marks values highlighted in the text; all reported p-values are two-sided.

5.1 Code-Mixing Penalty at Retrieval (H04)

Success@1 for each system on the deployed Hinglish queries and their gold English counterparts over all 3,015 pairs is reported in Table 2 and visualised in Fig. 1. Every system retrieves significantly worse on Hinglish at rank 1 (McNemar, largest $p = 0.018$), and the penalty is positive in all 28 system–metric combinations — four systems $\times$ Success@1/3/5/10, MRR@10, nDCG@10 and graded nDCG@10 — and significant in 26 (Fig. 2). Because the penalty holds across lexical, dense, and hybrid methods and at every depth, it is a property of code-mixing itself rather than of any single configuration, which is why the test is run per system. H04 is rejected for every system tested.

Table 2: Success@1, Success@10 and MRR@10 by retrieval system and query language ($n = 3,015$ pairs; higher is better). Penalty = English – Hinglish at rank 1; McNemar’s test per system. Bold indicates the best deployed-condition (Hinglish) system and the greatest penalty. Penalty 95% CIs (bootstrap): LaBSE [+0.0040, +0.0375], BM25 [+0.0743, +0.1085], Hybrid [+0.0036, +0.0408], TF-IDF [+0.0541, +0.0839]. Penalties at every depth and metric are shown in Fig. 2.

System	S@1 Hi	S@1 En	Penalty (En – Hi)	McNemar p	S@10 Hi	S@10 En	MRR@10 Hi	MRR@10 En
Hybrid (RRF)	0.1751	0.1973	+0.0222	0.018	0.6564	0.7393	0.2987	0.3438
LaBSE (passages)	0.1280	0.1486	+0.0206	0.017	0.6325	0.6783	0.2584	0.2835
BM25	0.0935	0.1847	+0.0912	9.9×10^{-26}	0.6116	0.7373	0.2173	0.3385
TF-IDF	0.0842	0.1529	+0.0687	1.9×10^{-19}	0.5599	0.6153	0.1972	0.2729
Random floor	0.0626	0.0626	—	—	0.4739	0.4739	—	—

The damage is sharply asymmetric. BM25 loses 0.0912 Success@1 to code-mixing against 0.0206 for dense LaBSE, roughly 4.4 times more, producing a crossover: BM25 is the best system on English queries (0.1847) and the worst on Hinglish (0.0935). The asymmetry widens with depth (Fig. 2): at rank 10 BM25’s penalty is +0.1257 against +0.0458 for LaBSE, and on graded nDCG@10 LaBSE shows no significant penalty at all (+0.0016, $p = 0.20$) while BM25 loses 0.0822 ($p = 7.5 \times 10^{-140}$). This crossover is the paper’s empirical argument for cross-lingual embedding at the retrieval layer: surface-form overlap is precisely what romanised code-switching removes.

On the single-vector index, MuRIL, an encoder pretrained on Indian languages with Hindi in Devanagari script, scores 0.0640 on Hinglish against the 0.0626 random floor: chance. On the same content in English it recovers to 0.1821. The failure is consistent with script mismatch rather than language mismatch: MMCQS is romanised, and the encoder has no purchase on the romanised surface. Transliterating the romanised queries into Devanagari tests that reading directly. MuRIL rises from 0.0640 to 0.0716 on Success@1 (not significant, $p = 0.24$) and improves reliably on rank-sensitive metrics (MRR@10 +0.0148, $p = 0.0096$; nDCG@10 +0.0077, $p = 0.0008$), yet recovers only 6.5% of the romanised-to-English gap. Script mismatch is therefore real but secondary: because the transliteration is a hand-verified mapping rather than a trained model this is a lower bound, but on this evidence writing the query in the encoder’s own script does not restore its English performance.

For comparison, Hinglish Success@1 on that index is 0.1343 for BM25, 0.1303 for multilingual-e5-base, 0.1167 for TF-IDF, and 0.1144 for LaBSE. Practitioners deploying Indian-language encoders on romanised user text should expect near-chance retrieval.

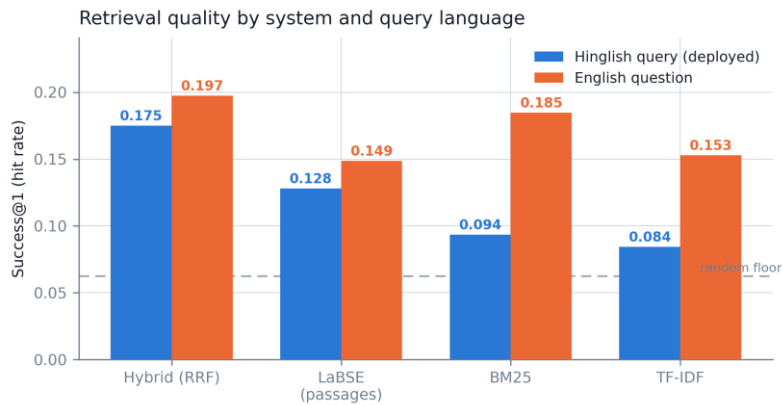


Fig. 1 Success@1 by retrieval system and query language over 3,015 pairs (higher is better). The dashed line marks the random floor (0.0626). Values correspond to Table 2.

Translation is the other query-side remedy, and it separates language from register. On a stratified subset of 500 queries translated into English by gpt-oss-120b, dense retrieval improves (LaBSE 0.1220 $\rightarrow$ 0.1680, $p = 0.040$) but lexical retrieval collapses (BM25 0.1080 $\rightarrow$ 0.0240, $p = 5.7 \times 10^{-8}$), below its Hinglish score. The mechanism is measurable: under half of a Hinglish query’s tokens exist in the English corpus at all, but those that do are rare and specific (mean IDF 5.22), because the Hindi carries the grammar and the English carries the clinical content; a fluent translation restores common function words (mean IDF 2.28) that dilute the lexical query. The same diagnostic bounds the main comparison: the gold English question averages 21 words against 97 for the translation, so it is a clinician-style summary, and the penalty in Table 2 compares a patient narrative with a clinical summary, conflating language with conciseness and register. H04 is unaffected — the penalty holds for every system and depth, and the MuRIL result is independent of it — but the estimate is an upper bound on the language effect alone; the translated arm, which holds length and register fixed, is the cleaner language-only comparison, and against it dense retrieval improves while lexical retrieval does not.

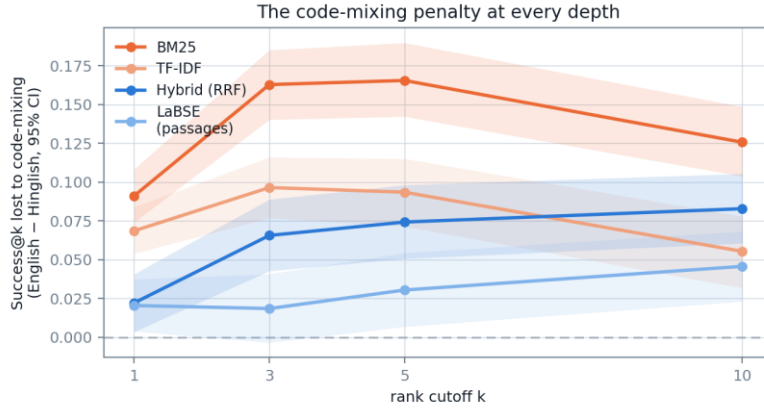


Fig. 2 The code-mixing penalty (English – Hinglish Success@k, 95% bootstrap CI) at every rank cutoff, per system. The lexical/dense gap widens with depth; a single rank-1 cutoff understates it.

5.2 Validity of the Scoring Instrument

In Table 3, the concept-precision instrument is audited against the unbiased reference before any generation result, using degenerate baselines ($n = 1,876$ for the constants). A hard-coded one-word answer, the constant string “swelling”, scores 0.7132, beating the real grounded system (0.1528) by a factor of 4.7.

The reason is structural: precision does not have a recall term, so its optimum is terseness. And so, absolute concept-precision levels therefore carry no information about answer quality; only paired differences between arms scored exactly the same are interpretable. As such, F_1 is reported wherever quality is at issue, and the baselines are delivered with the scored tables.

Table 3: Concept precision of degenerate baselines and of the two systems against the unbiased reference ($n = 1,876$ for the constants, 1,705 for the verbatim-copy ceiling, and 701 for the two systems; rows are dropped where an arm asserts no scoreable concept). Absolute levels are not quality measures: a one-word constant outcores the deployed system 4.7-fold.

Scored answer	Concept precision
Copy of the reference, verbatim	1.0000
Constant “swelling”	0.7132
Constant “swelling and erythema”	0.6586
Constant “erythema”	0.6039
Constant “rash swelling pain erythema infection lesion”	0.2338
Grounded system	0.1528
Zero-shot system	0.1066
Constant “pain”	0.0032

The same audit closes a double-counting hazard: the commonly reported hallucination rate is exactly $1 - \text{precision}$ (verified $\min = \max = 1.0$), so a factuality improvement and a hallucination reduction of matching size are one finding, not two, and every result below is reported once.

The instrument was also audited against an independent reader. qwen/qwen3.8-27b, a model family used nowhere else in the study, labelled 120 sampled answers (60 per arm, stratified by code-mixing tertile) with the concepts each positively asserts about the patient, from the same 26-item vocabulary. Agreement with the regex extractor is moderate (Cohen’s $\kappa = 0.494$; extractor precision 0.412 and recall 0.686 against the reader), and the extractor over-attributes: it counts hedged and hypothetical mentions that the reader excludes. The over-attribution is unequal across arms — precision 0.480 on grounded answers against 0.358 on zero-shot answers — so it does not cancel in the paired difference. Because a spurious concept enlarges the denominator of factual support, the zero-shot arm is depressed more than the grounded arm, and the measured grounding benefit of Sect. 5.3 is inflated by an amount this audit bounds but does not determine. This is automatic cross-validation, not clinical validation (Sect. 6.1). The lexicon’s coverage is also bounded: 98.3% of references contain at least one concept, but 13.9% contain exactly one, and the most frequent terms it cannot represent are misspellings in the free-text captions that serve as the unbiased reference.

5.3 Grounding Effect (H_{01}) and Its Reference-Sensitivity

Under the evaluation protocol in common use, scoring both arms against the retrieved evidence, grounding improves factual support strongly and consistently (Table 4). The effect survives the lexicon repair of Sect. 5.2 and replicates across the two generator families under oracle evidence (llama-3.1-8b +0.2962, $d = 0.720$; gpt-oss-120b +0.2608, $d = 0.640$); under real retrieval on gpt-oss-120b it is much smaller (+0.0704, $d = 0.191$), so H_{01} is rejected under the evidence-referenced protocol, and the replication is a property of oracle evidence rather than of the deployed pipeline. Following Sect. 5.2, the arm means underlying Table 4 are components of the paired estimand Δ , not quality measures.

Table 4 Grounding effect ($\Delta = \text{grounded} - \text{zero-shot within query}$) under both metrics and both references: concept precision against the evidence the grounded arm was given, and concept F1 against that evidence and against the image description neither arm saw. Each cell gives Δ with BH-corrected significance (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s.), n and Cohen’s d . Under the original lexicon the llama oracle contrast was +0.2345 ($d = 0.576$, $n = 1,165$). Absolute arm means are not quality measures (Sect. 5.2).

Generator / evidence	Precision vs evidence	F1 vs evidence	F1 vs unbiased reference
llama-3.1-8b, oracle	+0.2962 *** $n = 669$, $d = +0.72$	+0.2029 *** $n = 651$, $d = +0.68$	+0.0620 *** $n = 613$, $d = +0.22$
gpt-oss-120b, oracle	+0.2608 *** $n = 214$, $d = +0.64$	+0.0926 *** $n = 207$, $d = +0.28$	−0.0207 n.s. $n = 189$, $d = −0.07$
gpt-oss-120b, real retrieval	+0.0704 ** $n = 221$, $d = +0.19$	−0.0320 * $n = 210$, $d = −0.12$	−0.0469 * $n = 200$, $d = −0.17$
gpt-oss-20b, oracle	+0.1966 *** $n = 140$, $d = +0.45$	−0.0067 n.s. $n = 134$, $d = −0.02$	−0.0217 n.s. $n = 133$, $d = −0.06$
gpt-oss-20b, real retrieval	+0.1071 ** $n = 138$, $d = +0.25$	−0.0698 ** $n = 130$, $d = −0.26$	−0.0574 n.s. $n = 130$, $d = −0.16$

The magnitude, however, depends on what is scored against (Table 4). Re-scoring the same generations by concept F1, BH-corrected, against the circular reference (the retrieved evidence) versus the unbiased reference (the image description, which neither arm saw) changes the picture: llama-3.1-8b with oracle evidence falls from +0.203 to +0.062; gpt-oss-120b with oracle evidence falls from +0.093 to −0.021 (n.s.); and gpt-oss-120b under real retrieval reads −0.032 circular and −0.047 unbiased, both significant. Across the five conditions the precision effect against the evidence reference is positive in all five; the F1 effect against the unbiased reference is positive in one.

This reference-sensitivity is a methodological caution attached to the finding, not a retraction of it: the grounding effect of Table 4 is real under its stated protocol, and the size a paper reports depends substantially on the evaluation design it chooses.

Mechanistically, on gpt-oss-120b grounding trades recall for precision: it decreases concept recall by 0.138 (BH $p = 0.00025$), resulting in a more conservative model; the llama arm moves the other way (recall $+0.077$, BH $p = 9.3 \times 10^{-7}$), so the trade is generator-specific, and whether that trade pays off depends on the generator. Oracle evidence scores higher than real retrieval by $+0.075$, but the difference is not significant on either generator ($p = 0.226$, $n = 76$; $p = 0.106$, $n = 125$), so the condition filter’s advantage over honest retrieval is not established at this sample size. One consequence of grounding is a change in refusal behaviour. For gpt-oss-120b ($n = 467$) the zero-shot arm virtually never declines (0.2%), while the grounded arms decline on 29.8% (oracle) and 33.8% (real retrieval) of queries, usually stating that the evidence concerns another patient; coverage drops from 92.9% zero-shot to 49.3% and 52.0% respectively. Retrieval’s top-1 was condition-correct in only 10.7% of these rows. Refusals assert no scoreable concept and vanish from a naive mean, which would make the system look healthiest exactly where it fails; coverage is therefore reported beside every factuality number.

5.4 Code-Mixing Robustness of Generation (H₀₂)

H₀₂ concerns the grounded-minus-ungrounded difference, and it is tested as planned: a two-way design with arm varying within query, whose arm $\times$ code-mixing interaction is the change in the paired gain across tertiles of Hindi proportion ($n = 1,165$). The two pre-specified tests disagree — one-way ANOVA on the gain $F = 3.03$, $p = 0.049$; Kruskal–Wallis $H = 5.11$, $p = 0.078$ — so both are reported and neither is chosen after the fact; on the continuous measure the gain rises with Hindi proportion (Spearman $\rho = +0.081$, 95% CI $[+0.023, +0.138]$, $p = 0.006$), from $+0.204$ in the lowest tertile to $+0.274$ in the highest. Evidence against H₀₂ is therefore borderline on the tertile tests and significant on the continuous one. The per-arm decomposition (Table 5) explains the direction: zero-shot factual support degrades significantly as code-mixing intensifies ($\rho = -0.116$, BH $p = 0.0003$), while the grounded arm is flat ($\rho = -0.001$, n.s.) and statistically equivalent to zero effect (90% CI $[-0.048, +0.049]$ within a ± 0.10 bound chosen after the data were seen); neither arm shows a significant hallucination trend. The gain grows because the ungrounded baseline falls, not because grounding improves. Under the originally shipped code-mixing measure the interaction is not significant (Kruskal–Wallis $p = 0.19$), a difference attributable to the lexicon repair described below.

Table 5 Code-mixing intensity (Hindi proportion, repaired measure) against outcome, $n = 1,165$. First row: the pre-specified test on the paired gain (continuous p ; tertile tests ANOVA $p = 0.049$, Kruskal–Wallis $p = 0.078$). Remaining rows: exploratory per-arm decomposition, BH-corrected.

Outcome	ρ	95% CI	p	Reading
Gain (grounded – zero-shot), pre-specified	$+0.081$	$[+0.023, +0.138]$	0.006	rises
Grounded factual support	-0.001	$[-0.058, +0.057]$	0.98	flat (equivalent)
Zero-shot factual support	-0.116	$[-0.171, -0.059]$	0.0003	degrades
Grounded hallucination	-0.022	$[-0.080, +0.036]$	0.59	flat
Zero-shot hallucination	$+0.042$	$[-0.018, +0.101]$	0.30	flat

The code-mixing measure itself required repair. The original index counted the English words “doctor” and “please” as Hindi; they occur in 68.2% and 35.7% of the 1,165 evaluated queries respectively (71.0% and 38.0% across all 3,015 queries), distorting measured intensity. Two hallucination effects reported under the original measure did not survive the repaired lexicon and are withdrawn. All correlations above use the repaired measure.

5.5 Evidence Provenance (H₀₃): Exploratory

Does authoritative case evidence beat general biomedical text? The candidate corpora differ enormously in topical coverage of the 18 condition groups (MultiCaRe 67.9%, MMedBench-English 16.4%, PubMedQA 2.1%), so an unmatched comparison would measure corpus topicality, not evidence provenance. Four matched corpora of 1,872 documents each are constructed: MultiCaRe case reports, PubMedQA abstracts, MMedBench English exam text, and a sentence-shuffled MultiCaRe control; gpt-oss-120b answers $n = 160$ queries under each condition.

H03’s primary outcome, answer quality, could not be tested adequately: pairwise F1 differences are small and none is significant, and the four-way omnibus rests on only 13 of 160 queries scoreable under all four conditions; with refusal rates of 76–88%, that joint event has probability near 0.1%. Moreover, most “scoreable” answers are also refusals — hedges that decline to answer yet name a concept (MultiCaRe: 59 of 63; Table 6) — so the F1 column rests largely on refusals, and requiring a genuine answer leaves no query usable under all four conditions. H03 is therefore neither rejected nor supported. As a secondary outcome on a different dependent variable, provenance significantly changes refusal behaviour (Cochran’s $Q = 9.09$, $df = 3$, $p = 0.028$; Table 6), and in the opposite of the expected direction: case reports, the “authoritative” source, provoke the most refusals (88.1%); exam text provokes the fewest (76.2%). A powered replication needs the low-refusal prompt already implemented and roughly 77 queries for 50 usable ones; at the refusal rates observed here the same design needs 616 queries (about 2.8 million generation tokens), which is why it was not run. This absence of evidence, not evidence of absence, and the result is provisional.

Table 6 Evidence-provenance conditions on four topically matched corpora of 1,872 documents each (gpt-oss-120b, $n = 160$ per condition). “Also refusal” counts scoreable answers that the refusal detector also flagged.

Evidence corpus	Refusal	Mean F1	n scoreable	Also refusal
MultiCaRe (case reports)	88.1%	0.3265	63	59
Shuffled MultiCaRe control	82.5%	0.3186	61	53
PubMedQA (abstracts)	79.4%	0.3476	70	53
MMedBench (exam text)	76.2%	0.2659	66	56

5.6 Negative and Corrected Results

A published heuristic does not transfer to this setting. The similarity-gap heuristic for adaptive context selection proposed for multimodal medical RAG [6] fires when adjacent retrieval neighbours differ by at least 0.248 in similarity. On this data it fires on 0 of 3,015 queries: the largest gap occurring anywhere is 0.109. Swept across thresholds against fixed- k selection at matched evidence budget, it wins 0 of 6 settings; where precision is flat (0.112–0.115) at every setting while recall falls monotonically. The similarity gap carries no relevance information here, and no claim is made that adaptive context selection works in this setting.

A configuration defect dominated the original retrieval comparison. First dense index was built with a 128-token sequence limit, truncating 99.9% of case narratives (median 300 tokens) and 60.0% of queries, while the lexical baselines read full documents, giving BM25 roughly 2.4 times more information. Correcting the limit and adding passage chunking raised dense Success@1 on Hinglish from 0.1144 to 0.1280 and reversed the dense-versus-BM25 ordering; raising the sequence limit alone reaches 0.1310, so passage chunking is adopted not because it maximises that figure but because it puts dense and lexical retrieval on matched content. The uncorrected comparison supported the opposite conclusion; it is reported because silent truncation defects are easy to ship and expensive to detect. The withdrawn hallucination effects are reported in Sect. 5.4.

6. Discussion

The retrieval findings are the study’s most decision-relevant. The code-mixing penalty is significant under every system tested, at every depth and on every metric, and roughly 4.4 times larger for lexical than for dense retrieval at rank 1, with a wider gap at depth, producing a best-on-English and worst-on-Hinglish crossover for BM25. Because the English arm is a clinician-written summary rather than a translation, the estimate is an upper bound on the language effect alone (Sect. 5.1). For deployments whose users write romanised code-switched queries the architectural implication is direct: the retrieval layer should be cross-lingual and dense, or hybrid, because the surface-form matching is exactly what romanisation destroys; translating the query first does not rescue lexical retrieval.

The MuRIL result points to script rather than language: an encoder pretrained on the right language in the wrong script retrieves at chance (0.0640 against a 0.0626 floor) and recovers to 0.1821 on identical content in English. But transliterating the query into Devanagari recovers only 6.5% of that gap, so script is not the binding constraint: either the encoder’s Hindi representations are weaker than its English ones on this clinical content, or a hand-built transliteration is too approximate to carry the content, and the study cannot separate the two. Transliteration-aware pretraining or romanisation normalisation remains the indicated remedy; this study tests the cheapest version of it and finds it insufficient.

Grounding is better described as a robustness and behaviour mechanism than a guaranteed quality gain. It stabilises factual support against code-mixing intensity, **shifts one generator toward precision over recall**, and converts silent confabulation into explicit refusal when the evidence does not fit. Whether that trade profits depends on the generator (the two families disagree in sign under the unbiased reference), **the size of any reported gain depends on the scoring reference, and part of the measured gain is an extraction artefact whose direction favours the result (Sect. 5.2)**. None of this retracts the effect; it bounds how it may honestly be reported.

The evaluation findings generalise beyond this system. Factuality should be scored with F1 rather than precision alone; degenerate baselines should accompany scored tables, since a one-word constant outscoring a real system 4.7-fold is otherwise invisible; **concept extractors should be audited against an independent reader, per arm, because an arm-dependent bias does not cancel in a paired difference**; refusal and coverage belong beside every factuality number, because refusals vanish from naive means precisely where systems fail; factuality and hallucination should not be reported as two findings when one is the complement of the other; references neither arm saw are preferable; **dose-response should be tested as it was planned, on the paired difference, with per-arm correlations as the decomposition**; and corpora should be topically matched before effects are attributed to provenance.

Finally, hosted-model decommissioning is a concrete reproducibility hazard: the generator behind the largest analyses disappeared mid-study. Archiving raw outputs kept every number in this paper re-scorable; without that discipline, the corrections reported here would have been impossible.

6.1 Limitations and Threats to Validity

Ten limitations bound the claims, and each comes with its measured extent. First, the relevance labels are coarse: condition-group matching over 18 groups admits same-group but clinically irrelevant cases, so absolute Success@1 is a lower bound on true relevance; **the graded variant reported in Sect. 4.2 is an automatic proxy, not a clinical judgment**. Second, the concept lexicon includes only 26 concepts and has not been validated by clinicians; **against an independent model reader its extractor reaches $\kappa = 0.494$ with precision 0.412, and the over-attribution is larger in the zero-shot arm, which inflates the grounding effect of Sect. 5.3 by an amount that is bounded but not determined. Clinician validation remains future work**. Third, the unbiased reference is narrow and repetitive: it averages 1.5 concepts, takes 412 distinct values across 2,988 rows, and a single value accounts for 22% of rows; cluster bootstraps deal with the induced non-independence, but the construct captured is partial, **and its free-text captions contain frequent misspellings that a lexical extractor cannot match**. Fourth, generation results are generator-dependent: two results differ in sign between the generator families under the unbiased reference, and both are reported with neither being chosen.

Fifth, provider instability is a real threat: llama-3.1-8b-instant, the generator behind the $n = 1,165$ analyses, was decommissioned mid-study and now returns HTTP 404, so its outputs remain archived and re-scorable but no new generations are possible, a genuine danger for research on hosted models; **the gpt-oss models are open-weight, so their results are regenerable in principle, and the archived generations make every number re-scorable**. Sixth, the H03 answer-quality comparison is under-powered, with only 13 of 160 queries scoreable under all four evidence conditions, **most of them refusals; a powered replication needs roughly 616 queries under the present prompt or 77 under the low-refusal one** (Sect. 5.5). Seventh, no clinical deployment claim is made: absolute Success@1 of 0.1751 on Hinglish, roughly two to three times the random floor, is far below what patient-facing use would require, **although Success@10 of 0.656 makes the pipeline usable as a first-stage candidate generator**. Eighth, the English arm of the penalty comparison is a clinician-written summary, not a translation, so the penalty is an upper bound on the language effect and conflates language with conciseness and register (Sect. 5.1). Ninth, the transliteration is a hand-verified lexicon with rule-based mapping for the tail rather than a trained model, so its recovery is a lower bound, and the translation baseline covers a 500-query stratified subset. Tenth, H04 was formulated after the initial retrieval experiments and is confirmatory only in the sense that it was then tested on the **full matched set**.

7. Conclusion and Future Work

This study set out to characterise, rather than improve, retrieval-augmented generation for Hinglish clinical questions over English case reports. Code-mixing significantly degrades clinical retrieval, about four times more for lexical than for dense methods **at rank 1 and more at depth; neither transliteration nor translation removes the penalty for lexical retrieval**; and while grounding **increases evidence-referenced factual support** and shields it from the code-mixing intensity, the apparent size of that benefit depends substantially on the evaluation protocol **and is partly an extraction artefact**.

The contributions are the retrieval-stage code-mixing penalty with its lexical/dense asymmetry and tested script-mismatch mechanism; pre-specified hypothesis tests replicated across two generator families under oracle evidence; and a demonstration that evaluation design changes the reported benefit, with the degenerate baselines and extractor audit needed to detect it.

Future work should replace coarse condition-group labels with graded clinical relevance, validate the 26-concept lexicon with clinicians (a blinded annotation protocol is released with the code), adopt richer references, replace the summary-based English arm with human translations so that language and register are separated, test transliteration-aware encoders against script mismatch, and power the provenance comparison for answer quality with the low-refusal prompt. Each tightens a bound measured here.

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Declarations:

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Data Availability: MMCQS [1] and MultiCaRe [2] are publicly available datasets. Scored result files and generator outputs are version-controlled in the code repository (see Code availability); derived pairing tables and FAISS indices are deposited at <https://doi.org/10.5281/zenodo.22166651>.

Materials Availability: Not applicable.

Code Availability: Available at <https://github.com/saaachis/hinglish-grounded-medical-rag>, release tag v1.1-iccsdi2026-revised. Archived outputs permit re-scoring of every reported number (Sect. 6.1), and `python -m src.analysis.verify_paper_numbers` checks every headline value in this manuscript against the committed results.

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